



human eye is much more sensitive to green light than red and blue, and has a much greater resolving power in that range.

The space between adjacent light-sensitive pixels is used to house on-chip electronics. Micro lenses placed above the filter help in directing light into adjacent pixels. As outlined earlier, a single pixel allows photons of one colour only. Full-colour image is worked out by using a complex method called demosaicing. Simply stated, the camera treats each 2x2 set of pixels as a single unit, thereby providing one red, one blue and two green pixels. The actual colour is estimated based on the photon levels in each of these four pixels.

Electron multiplication CCD

Electron-multiplying CCD (EMCCD) technology helps to overcome the shortcoming of traditional CCD technology in offering high sensitivity at high speed. Traditional CCD sensors backed by readout noise figures of typically less than ten electrons offer high sensitivity. High sensitivity, however, comes at the cost of reduced speed of readout, which is typically less than 1MHz. The speed constraint arises from bandwidth limitation of the CCD charge amplifier. Increasing bandwidth, which is essential for high speed operation, increases noise too. That is, high-speed amplifiers are relatively noisier than their slow-speed counterparts.

Electron-multiplying CCD overcomes this limitation by building a unique electron multiplying structure into the chip. As a result, EMCCD as an image sensor is capable of detecting single photon events without an image intensifier. EMCCD sensors achieve high-sensitivity, high-speed operation by amplifying charge signal before the charge amplifier.

Most electron-multiplying CCDs utilise a frame-transfer CCD structure, wherein sensor area captures the image and storage area stores the image prior to readout. Storage area is normally identical in size to sensor area and is covered with an opaque mask, normally made of aluminium. When sensor area is exposed to light,

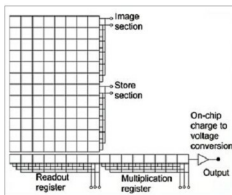


Fig. 6: Electron-multiplying CCD

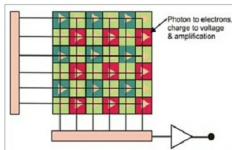


Fig. 7: CMOS sensor

an image is captured, which is automatically shifted downwards behind the masked region of the chip, and subsequently read out. During the readout process, another image is being captured by sensor area. The aluminium mask therefore acts like an electronic shutter.

As shown in Fig. 6, there is a multiplication register between the normal serial readout register and the charge amplifier. To read out the sensor, the charge is shifted out through the readout register and through the multiplication register where amplification occurs prior to readout by the charge amplifier.

The multiplication register has several hundred stages or cells that use higher-than-normal clock voltages to achieve charge amplification. Amplification occurs in the multiplication register through a process known as clock-induced charge or spurious charge that occurs naturally in CCDs.

When clocking the charge through a register, there is a very small but finite probability (typically less than 2 per cent) that the charges being clocked can create additional charges by a process known as impact ionisation. Impact ionisation is the process by which a charge

having sufficient energy creates another electron-hole pair. Hence a free electron charge in the conduction band can create another charge, leading to amplification.

Electron multiplication factor (M) may be computed from:

$$M = (1 + p)^N$$

where N is number of cells and p is the probability value. For instance, if the multiplication register had 512 cells or stages, and the probability of secondary electron generation were 1.3 per cent, the multiplication factor would be around 744. Electron multiplication prior to the output amplifier ensures that the readout noise introduced by the amplifier has negligible effect.

Major advantages of EMCCD sensors include high sensitivity in low light conditions, high-speed imaging capability, good daytime imaging performance and reduced likelihood of sensor damage while viewing in bright conditions. By elevating photon-generated charge above the readout noise of the device, even at high frame rates, the EMCCD meets the needs of ultra-low-light imaging applications without the use of external image intensifiers. The disadvantage is its relatively higher power consumption due to need for active cooling of the CCD.

The extreme low light capability of these EMCCDs enables a range of applications including border and coastal surveillance, surveillance of ports and airports, protection of sensitive sites and critical assets, and low-light scientific imaging such as in astronomy.

CMOS sensors

For their operation, both CCD and CMOS sensors depend on photoelectric effect to create electrical signal from light photons. These are two different technologies for digital image capture. Each has unique strengths and weaknesses for different applications.

In a CCD sensor, the quantum of charge held by different pixels is transferred through one or a very limited number of output nodes. Each pixel's charge is converted into a proportional voltage and, after buffering,